

EXTERNAL SESSION-BASED GEOMETRY SCANNER — SCAN MODES, VALIDATION, QUALITY INDICES, AND ANTI-SPOOF (NON-LIMITING)

A. Scan Session Workflow (Non-Limiting)

In various embodiments, the geometry scanner is used in a discrete scan session configured to produce a baseline geometry reference with confidence indicators. A non-limiting workflow includes: 1. initial placement of the C-shaped or closed-ring scanner at a starting position; 2. contact confirmation using pressure/contact sensing, capacitive confirmation where available, and/or optical coupling checks; 3. scan motion guidance, wherein the user performs a guided scan along a direction while the device acquires sensor streams; 4. segment validation, wherein windows of data are evaluated for quality and plausibility; 5. baseline generation, wherein valid segments are used to compute baseline length and diameter references; and 6. repeatability scoring, wherein multiple scans may be compared to compute a repeatability index and confidence indicator. In certain embodiments, the computing system provides guidance cues to slow down, re-seat, or repeat the scan when quality indices indicate low confidence.

B. Length Estimation Using Optical Flow (Non-Limiting)

In various embodiments, one or more optical flow sensors are used to estimate incremental displacement along a scan path. The system may: 1. integrate incremental motion estimates to compute an estimated scan length; 2. fuse optical flow with IMU-derived motion estimates to reduce drift; and/or 3. reject segments where optical flow indicates loss of coupling, excessive slip, or inconsistent motion. In certain embodiments, multiple optical flow sensors are used to improve robustness and reduce sensitivity to localized surface variation.

C. Diameter Estimation Using Distance Sensors (Non-Limiting)

In various embodiments, diameter estimation is performed using one or more distance sensors including ToF, LiDAR, and/or IR distance sensors. Non-limiting approaches include: 1. measuring distance from one or more sensor regions to tissue-adjacent surfaces; 2. combining multiple distance measurements around the inner circumference to estimate diameter and/or cross-sectional profile; and 3. applying plausibility constraints and smoothing to reduce transient measurement noise. In certain embodiments, distance sensors are time-multiplexed to reduce cross-interference and are compensated for ambient conditions.

D. Temperature Context and Drift Compensation (Non-Limiting)

In certain embodiments, the geometry scanner includes one or more temperature sensors configured to provide thermal context and/or compensate temperature-dependent drift in sensor behavior. Non-limiting uses include: 1. confirming on-body coupling via plausible thermal trends; 2. gating segments when temperature indicates poor coupling; and/or 3. compensating distance sensor drift and/or optical sensor drift using temperature-dependent correction factors.

E. Quality Indices and Validity Gating (Non-Limiting)

In various embodiments, the computing system computes one or more quality indices (QI) to accept, down-weight, or exclude segments. Non-limiting QI components include: 1. contact integrity index, derived from pressure/contact sensing stability and/or coupling continuity; 2. motion stability index,

derived from IMU-based jerk, rotation, and orientation stability; 3. optical flow coherence index, derived from continuity, saturation, and plausibility checks in optical flow output; 4. distance sensor plausibility index, derived from range bounds, discontinuity detection, and cross-sensor agreement; 5. thermal plausibility index, derived from temperature coupling dynamics; and 6. multi-sensor agreement index, derived from coherence between optical flow, IMU, contact sensing, and distance sensing. In certain embodiments, a segment is considered valid only when it meets thresholds across at least two indices including motion stability and at least one of contact integrity, optical coherence, or distance plausibility.

F. Anti-Spoof and Physiologic Confirmation (Non-Limiting)

In certain embodiments, the system incorporates anti-spoof measures including: 1. requiring contact confirmation via pressure/contact sensing; 2. requiring multi-sensor agreement between optical flow, IMU motion, and distance sensor behavior; and/or 3. using PPG-derived pulse features as a physiologic confirmation signal for at least a subset of the scan session when available. In certain embodiments, the computing system generates an anti-spoof confidence score and gates baseline outputs such that low-confidence scans are labeled reduced confidence or require re-scan.

G. Privacy-Preserving Outputs (Node 3) (Non-Limiting)

In various embodiments, Node 3 outputs are presented without explicit anatomical imagery and without explicit anatomical labeling. Outputs may include baseline length indices, diameter indices, zone-based profile summaries, repeatability indices, and confidence indicators.